

## Problem Set for 07 April 2025

### Example (Quotient Spaces in Analysis).

1. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be of period  $T$ . Then there exists a unique function  $\tilde{f} : (\mathbb{R}/T) \rightarrow \mathbb{R}$  mapping the coset  $[x] := x + T \cdot \mathbb{Z}$  to  $f(x)$ . Explicitly,

$$\tilde{f} : (\mathbb{R}/T) \rightarrow \mathbb{R}, \quad (x + T \cdot \mathbb{Z}) \mapsto f(x).$$

2. The derivation  $\frac{d}{dx} : D(\mathbb{R}) \rightarrow \mathbb{R}^{\mathbb{R}}$  maps differentiable functions  $f$  to  $f'$ . There exists a unique linear map

$$D(\mathbb{R})/\mathbb{R} \rightarrow \mathbb{R}^{\mathbb{R}}, \quad (f + \mathbb{R} \cdot \text{id}) \mapsto f'.$$

3. Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a function satisfying  $f(x, y) = f(x - t, y - t)$  for all  $x, y, t \in \mathbb{R}$ . Then  $f$  is automatically defined on  $\mathbb{R}^2/\text{span}((1, 1)) \simeq \mathbb{R}$ .

4. etc.

**Exercise** Let  $V$  be a vector space over  $\mathbb{F}$ , and define  $\text{End}(V) := \text{Hom}_{\mathbb{F}}(V, V)$ . Define:

- $f \in \text{End}(V)$  as *almost zero* (denoted  $f \in \text{AZ}(V)$ ), provided  $\dim \text{im}(f) < \infty$ .
- $f \in \text{End}(V)$  as *almost an isomorphism* (denoted  $f \in \text{AI}(V)$ ), provided  $\dim \ker f + \dim \frac{V}{\text{im}(f)} < \infty$ .

Now finish the following.

1. If  $\dim V < \infty$ , determine  $\text{AZ}(V)$  and  $\text{AI}(V)$ .
2. If  $V = \mathbb{R}[x]$ , find an injective  $f \in \text{AI}(V)$  and a surjective  $g \in \text{AI}(V)$ . Additionally, find some  $h \notin \text{AI}(V) \cup \text{AZ}(V)$ .
3. Determine whether  $\text{AZ}(V)$  is a subspace of  $\text{End}(V)$ . What about  $\text{AI}(V)$ ?
4. Show that for arbitrary  $f, g \in \text{AZ}(V)$ , one has  $g \circ f \in \text{AZ}(V)$ .
5. Prove that  $\text{AZ}(V)$  is an **ideal** in  $\text{End}(V)$ . The term **ideal** is defined in 20 March 2025.
6. Show that for arbitrary  $f, g \in \text{AI}(V)$ , one has  $g \circ f \in \text{AI}(V)$ .
7. Demonstrate that if any two elements of  $\{f, g, g \circ f\}$  belong to  $\text{AI}(V)$ , then so does the third.

*Hints: Show that*

1.  $[\ker g \circ f \rightarrow \ker g, \quad x \mapsto f(x)]$  has kernel  $\ker f$ ;
2. the dual statement of  $\uparrow$ ;
3. for  $f \in \text{AI}$ , set  $\Phi(f) := \dim \ker f - \dim \frac{V}{\text{im}(f)}$ , show that  $\Phi(f \circ g) = \Phi(f) + \Phi(g)$ .

8. Show that for arbitrary  $f \in \text{AZ}(V)$ , one has  $(\text{id}_V + f) \in \text{AI}(V)$ .
9. **(Optional)** By Axiom of Choice, every subspace has a complementary subspace. Prove that  $f \in \text{AI}(V)$  can always be expressed as the summation of a map of  $\text{AZ}(V)$ -type and an automorphism (i.e., auto = iso + endo). Conversely, show that any such summation belongs to  $\text{AI}(V)$ .
10. **(Optional)** Show that  $\frac{\text{End}(V)}{\text{AZ}(V)}$  is a quotient ring with a well-defined subset  $\frac{\text{AI}(V)}{\text{AZ}(V)}$ , consisting precisely of the invertible elements. In this quotient ring, the operators  $\frac{d}{dx}(-)$  and  $\int_0^x (-) dx$  are mutually inverse.

**Problem (Optional)** Let  $V$  be a vector space over  $\mathbb{F}$ . We define

$$\prod_{i \in I} V := V \times \underbrace{\cdots \times}_{I\text{-copies}} V \simeq \text{Hom}_{\text{Sets}}(I, V)$$

as an  $\mathbb{F}$ -vector space whose elements are sequences  $(v_i)_{i \in I}$ . Informally,  $\prod_{i \in I} V$  represents the space of sequences indexed by  $I$  with values in  $V$ .

1. Define  $e_i := (\dots, 0, 0, \underbrace{1_{\mathbb{F}}}_{i\text{-th entry}}, 0, 0, \dots) \in \prod_{i \in I} \mathbb{F}$ . Determine when  $\{e_i\}_{i \in I}$  forms a basis of  $\prod_{i \in I} \mathbb{F}$ .
2. Describe  $\text{span}(\{e_i\}_{i \in I})$  in natural language.
3. Substituting  $I$  with  $V_{\text{as a set}}$ , the underlying set of  $V$  as an  $\mathbb{F}$ -vector space, prove that  $\pi$  is a linear surjection and determine the equivalence relation  $R_{\mathbb{F}}$  as defined in 24 March 2025:

$$\pi : \text{span}(\{e_v\}_{v \in V_{\text{as a set}}}) \rightarrow V, \quad e_v \mapsto v.$$